

Vatsal Agrawal

Full-Stack Developer · AI/ML Engineer · Automation Specialist
Second-year CS undergrad who ships — AI, mobile, web, and automation.

CONTACT

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EDUCATION

B.Tech — CSE (AI & ML)

VIT Chennai ·
2024–2028

LANGUAGES

Python · C/C++ · Java
JavaScript · SQL /
PL-SQL
HTML · CSS

FRAMEWORKS

React · Flutter ·
Streamlit
TensorFlow · LangChain
Pygame · Ursina

AI & AUTOMATION

Gemini API · Google ML
Kit
fal.ai · NEAT Algorithm
n8n · Prompt Engineering
OCR · Document AI

TOOLS & CONCEPTS

PROFESSIONAL PROFILE

Versatile Computer Science undergraduate at VIT Chennai (AI & ML specialisation) with a demonstrated ability to independently conceive, build, and ship production-ready software across five domains: AI/ML engineering, full-stack web development, cross-platform mobile apps, game development, and intelligent automation. A self-driven builder who thrives at the intersection of disciplines — applying ML to games, AI to healthcare, and automation to content creation.

FEATURED PROJECTS

MockMate²⁰²⁶

Flutter · Gemini API · Google ML Kit · Computer Vision

- Multimodal AI interview coach — generates dynamic questions from user CV/JD via Gemini API, records video responses, and uses ML Kit to analyse facial expressions and vocal confidence, delivering structured improvement reports.

InsuTrack²⁰²⁶

Flutter · Python · LangChain · Streamlit · OCR

- Collaborative dual-portal healthcare platform with real-time doctor–patient communication, prescription tracking, and a peer-support community. Includes a deployed LangChain OCR SaaS that consolidates multi-prescription histories into clinical summaries.

Autonomous Content Pipeline²⁰²⁵

n8n · YouTube API · Google Sheets API

- Zero-touch AI content pipeline: selects video concepts from Google Sheets → generates AI taglines → embeds onto video → publishes to YouTube daily. Fully automated from concept to distribution.

Self-Learning Flappy Bird AI²⁰²³

Python · TensorFlow · NEAT Algorithm

- NEAT neuroevolution agent that masters Flappy Bird through genetic selection across generations —

Git · REST APIs
Data Structures &
Algorithms
OOP · System Design

SOFT SKILLS

Leadership · Prototyping
Creative Problem Solving
Cross-domain
Development
Self-directed Learning
Typing Speed: 120 WPM

HACKATHONS

FlutterSprint — GDG
🏆 2nd Place, Built
MockMate, an
AI-powered mock
interview app, under this
hackathon organised by
Google Developers Group.

emergent perfect play with no hardcoded strategy.
Demonstrates applied ML and evolutionary algorithms.

Space Invaders Arcade Game²⁰²³

Python · Pygame

- Feature-complete arcade game with home screen, multi-level difficulty, coin economy, and item shop — built from scratch with a clean, decoupled game-loop architecture.

ADDITIONAL PROJECTS

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- 3D Obstacle Course Game — third-person 3D game built with Python's Ursina engine; full physics, collision, and level progression.
 - Thumbnail Creator SaaS — generates four AI thumbnail variants (16:9 & 9:16, standard & clickbait) from a YouTube URL.
 - LinkedIn Post AI Agent — fully automated n8n pipeline for AI-written and published LinkedIn content.
 - A* Pathfinding Visualiser — interactive A* algorithm implementation with priority queues and real-time step rendering.
 - Dijkstra Pathfinding Visualiser — multi-destination shortest path visualiser with real-time exploration and full distance breakdown via Tkinter.
 - Multiplayer Game Hub (In Progress) — browser-based platform for party voice calls, game voting, and real-time multiplayer.